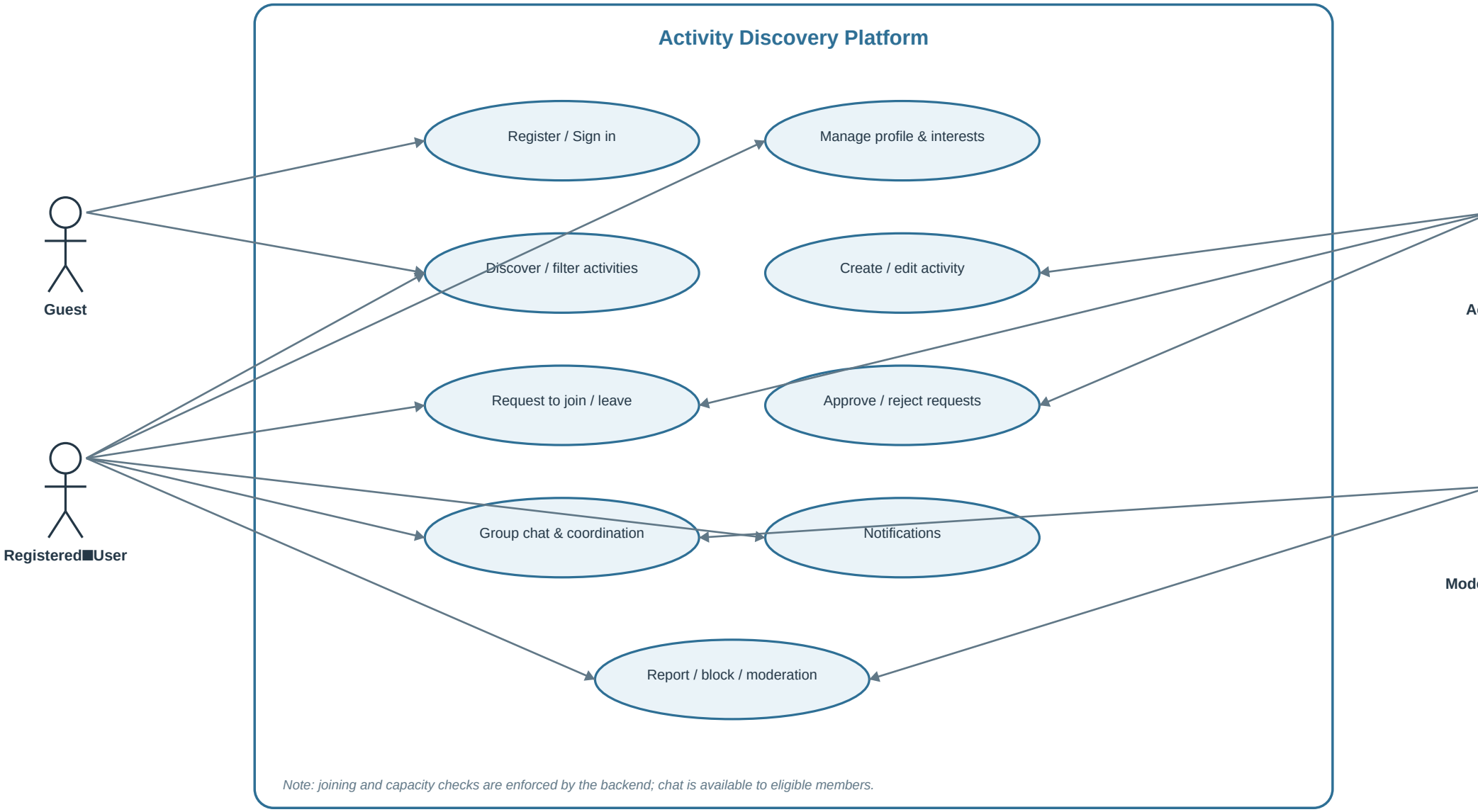
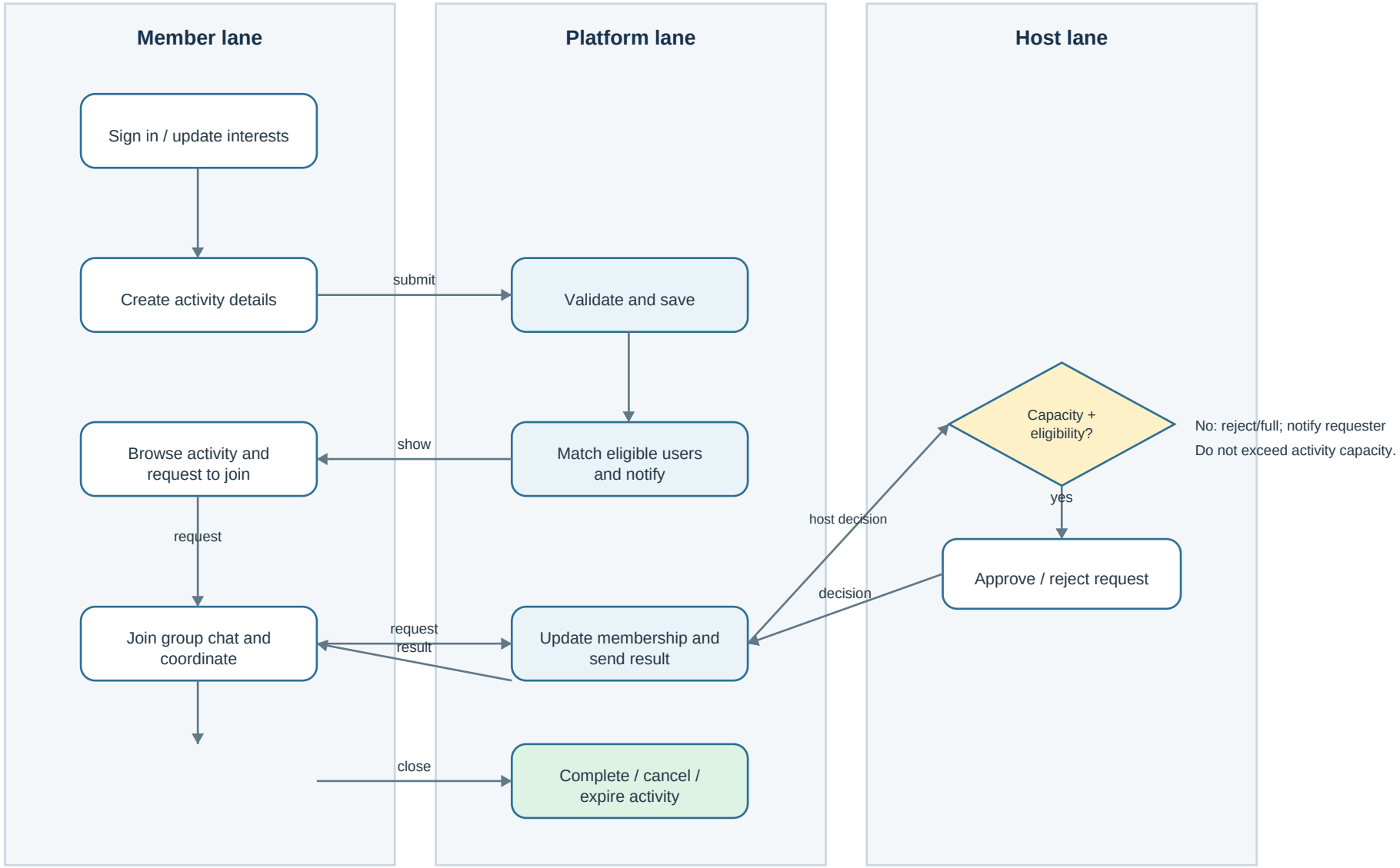
1. Use-Case Diagram

Actors and major capabilities of the group activity discovery platform



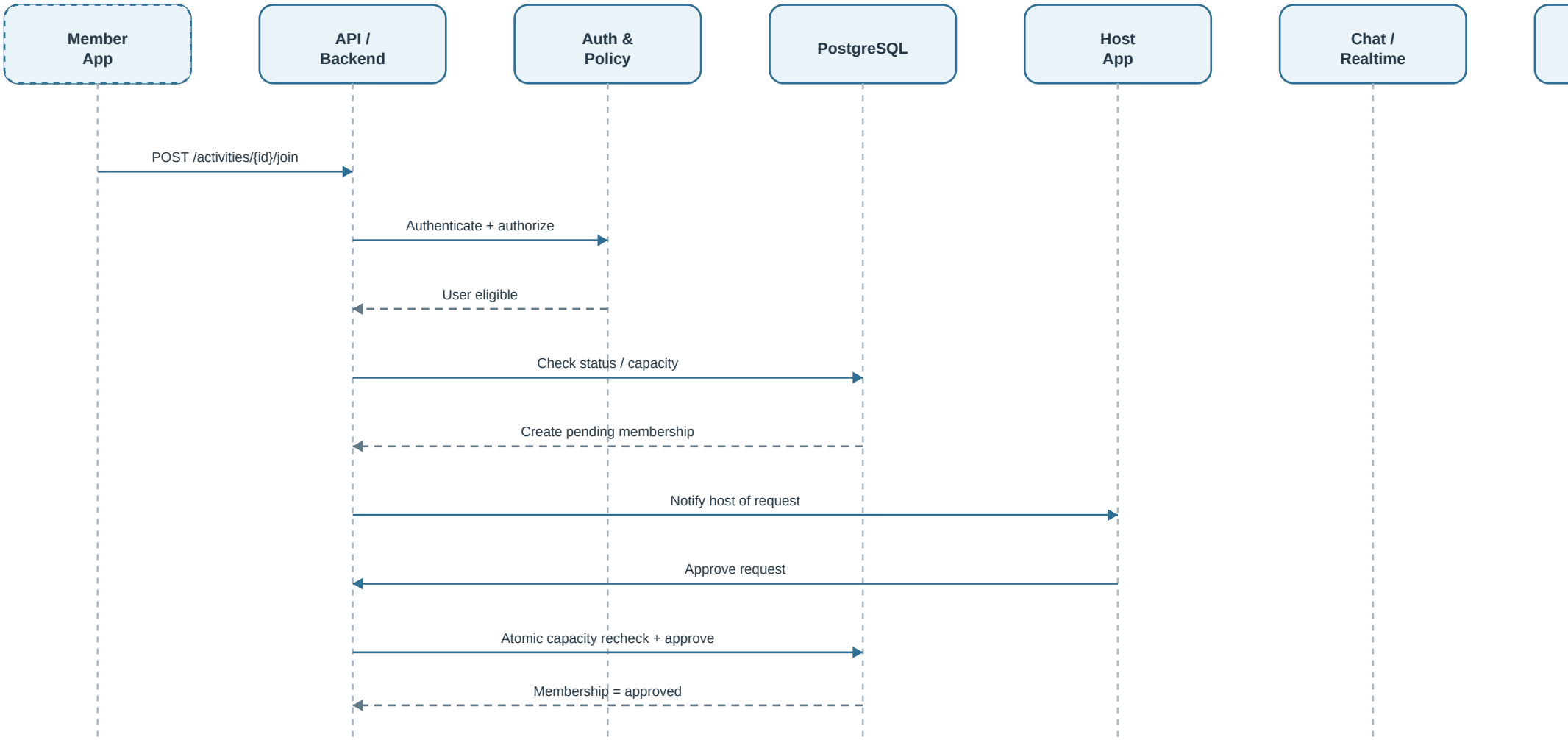
2. Activity Diagram

End-to-end workflow: create an activity, discover it, join, coordinate, and close



3. Sequence Diagram

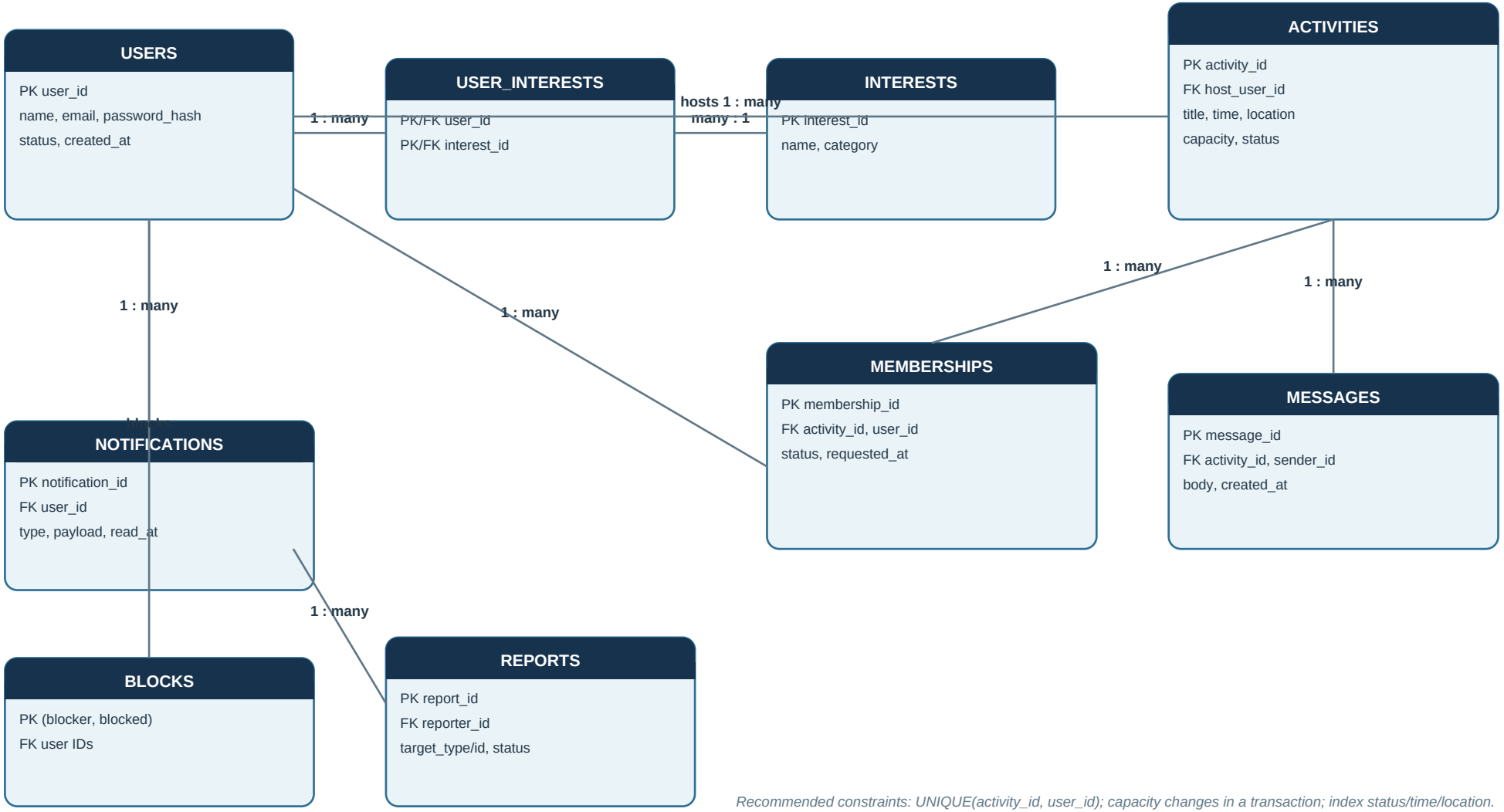
Successful join request with host approval and membership creation



Backend publishes membership change → chat access enabled → participant and host receive notifications.

4. ER Diagram (Conceptual)

Core data entities and cardinalities for discovery, membership, chat, and safety

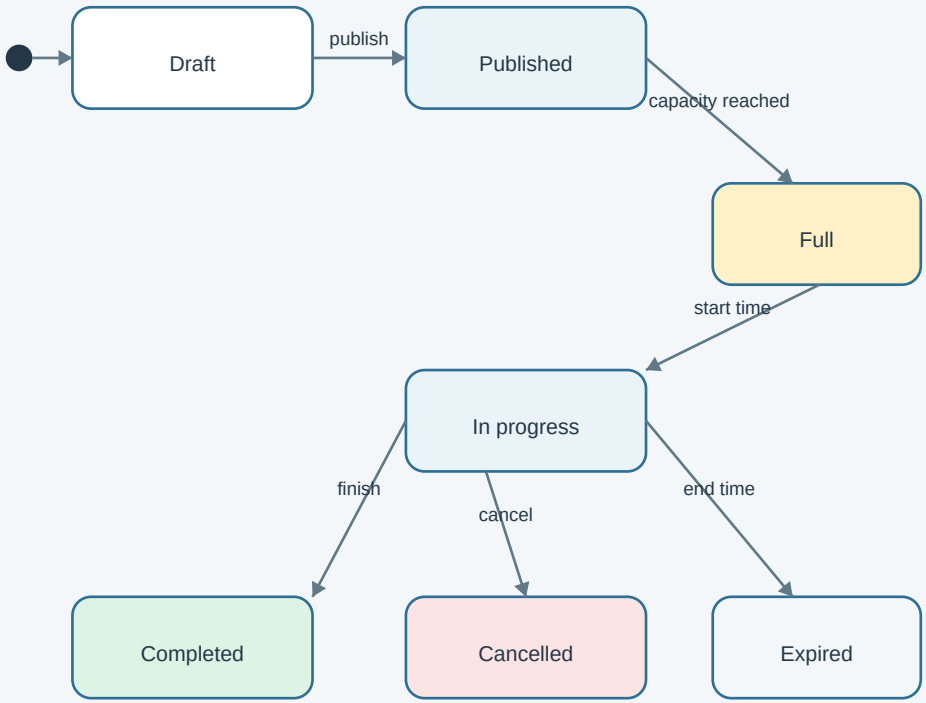


Recommended constraints: UNIQUE(activity_id, user_id); capacity changes in a transaction; index status/time/location.

5. State Diagram

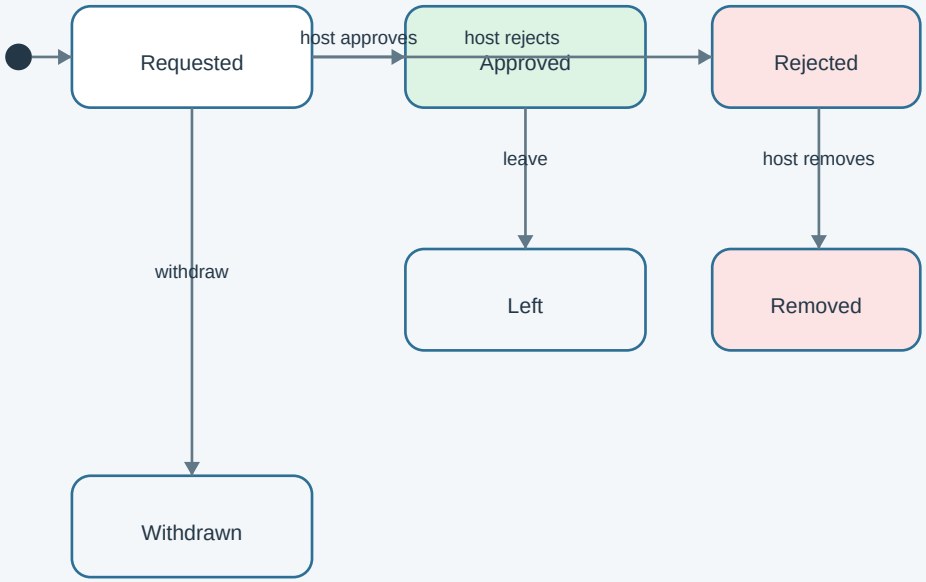
Activity lifecycle and join-request membership lifecycle

Activity lifecycle



Only the host can approve/reject join requests. Activity status and membership status are separate.

Membership request lifecycle



Approval must re-check capacity and blocking/eligibility rules on the server.